F1+TF1 Computer Networks: Case Study Review 3

1. Case Study Title

CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks

Team Details

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2. Software Implementation Overview

This review documents the software-only CoWiC-MAC simulation built for the case study. The implementation follows Cheng, Li, and Pan (*Computer Networks*, 2026, Art. 112479).

Stack:

Layer	Choice
Language	Python 3.11+
Discrete-event engine	SimPy
MILP	PuLP + CBC (`pulp < 3.0.0`)
Tests	pytest
Lint / types	ruff, mypy

Verification status at document time:

- 245 automated tests passing
- CI green on Python 3.11 and 3.12
- Clean `mypy` and `ruff` checks

Implemented modules:

Area	Paths
Models	`src/cowic/models/frame_group.py`, `src/cowic/models/superframe.py`
WiFi	`ap_orchestrator.py`, `milp_scheduler.py`, `baselines.py`, `fallback.py`
ZigBee	`coordinator.py`, `lwsa_adapter.py`
CTC	`src/cowic/ctc/channel.py`
Simulation	`simulation_engine.py`, `channel_medium.py`
Analysis	`metrics.py`, `plotter.py`
Traces	`traces/synthetic_generator.py`, `traces/playback.py`
Sweep	`scripts/run_benchmark_sweep.py`

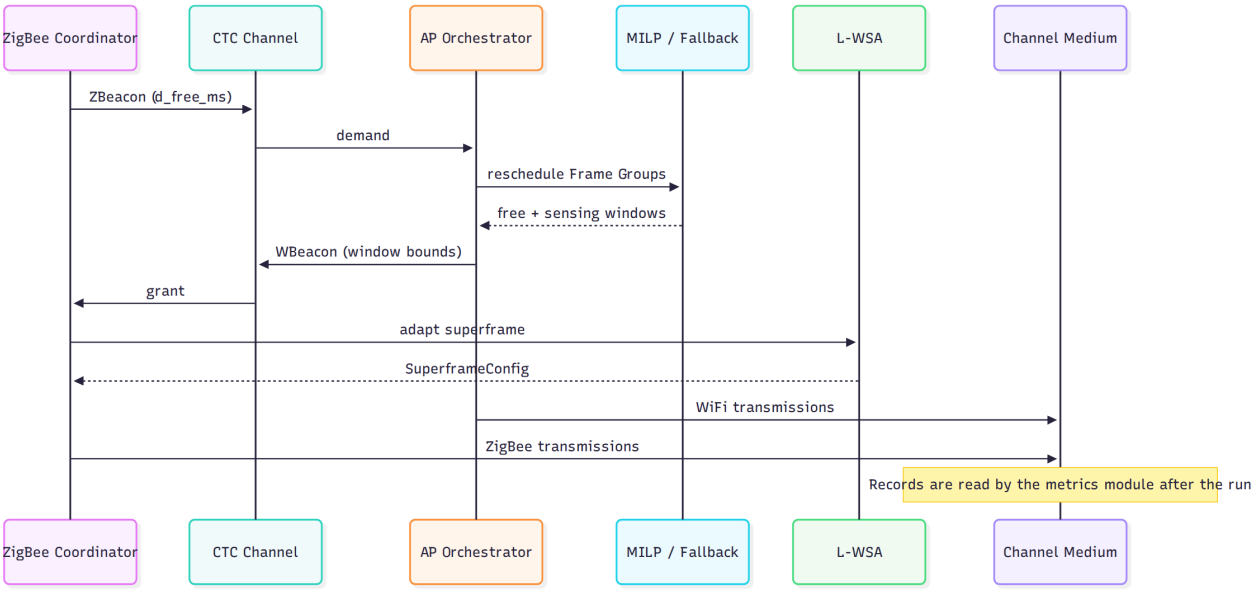
3. Contributor Workstream Summary

Presentation ownership matches Review 2.

Contributor	Presentation domain	Main implementation focus
Upayan Mazumder	Central architecture	Simulation engine, CTC, channel medium, MILP scheduler, L-WSA, plotter, integration
Nitesh Pandey	WiFi side	Trace generator, AP orchestrator, baselines, fallback, hard-constraint MILP (Exp 5), engine fixes, metrics, benchmark sweep
Divyansh Saxena	ZigBee side	ZigBee architecture review, superframe / L-WSA specification input, test design

4. System Architecture and Data Flow

The system has three domains: WiFi, ZigBee, and the shared 2.4 GHz medium. CTC joins the two MAC domains.



CTC payloads:

Message	Direction	Fields
ZBeacon	ZC → AP	`d_free_ms`
WBeacon	AP → ZC	`t_free_start_ms`, `t_free_end_ms`, `t_sense_start_ms`, `t_sense_end_ms`

Residual WiFi fragment lists stay outside the WBeacon scalar payload. The ZC receives them as a separate input, matching the base-paper model where the ZC senses leftover WiFi activity inside the granted window.

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5. WiFi Frame Group MILP Formulation

Objective (Equations 11–17):

$$\min \sum_{k=1}^{K} z_k^{\text{free}} d_k + \gamma \sum_{k=1}^{K} z_k^{\text{sense}} d_k + \lambda \sum_{k=1}^{K} y_k + \sum_{k=1}^{K} \mu_k P_k$$

Constraint groups:

1. Delay definition for y_k
2. Causality with respect to arrival times
3. Deadline and cycle-end bounds
4. Feasibility of postponed Frame Groups in the next cycle
5. Inter-FG non-overlap through Big-M ordering binaries
6. Overlap indicator constraints for free and sensing windows
7. Hard sensing-window protection: $\sum_k z_k^{\text{sense}} = 0$

Solver path: PuLP model solved with the CBC solver that PuLP includes. A HiGHS solver can be supplied as an option. If status is not optimal, the AP uses whitespace expansion fallback.

Postponed FG reporting convention: a postponed FG is reported at `scheduled_start_ms = t_end_ms` with `delay_ms = max(0, t_end_ms - original_start_ms)`.

6. ZigBee L-WSA Superframe Adaptation

Objective:

$$\eta_{\text{total}} = m \cdot t_{\min} \cdot \eta_{\text{cap}}^{p} + \alpha \cdot n \cdot t_{\text{SfSlot}} \cdot \eta_{\text{cfp}}^{p}$$

Search variables and domains:

Variable	Domain
t_{SfSlot}	$0.96 \cdot 2^{SO}$ ms, $SO \in [0, 8]$
j	$[0, 7]$
i	$i + j \leq 15$
t_{offset}	window start and residual-fragment boundaries

Filter conditions:

- Active superframe duration must fit the free window.
- CAP slots contribute only when they clear a $t_{\min}$ occupancy check.
- CFP slots contribute only when the full slot clears residual WiFi.

IEEE 802.15.4 2.4 GHz O-QPSK constants used:

Constant	Value
Symbol duration	0.016 ms
Base slot	0.96 ms
Backoff slot	0.32 ms
Turnaround	0.192 ms
ACK	0.352 ms
Max packet	4.064 ms

7. Cross-Technology Communication Model

The base paper uses WiCast for WiFi→ZigBee downlink and SymBee for ZigBee→WiFi uplink. This project does not emulate those waveforms.

Instead, CTC is a discrete-event control channel:

- Message types: `WBEACON`, `ZBEACON`
- Configurable packet error rate
- Scalar float payloads only

This abstraction is enough for MAC-layer coordination studies. It is not a PHY fidelity claim.

8. Experimental Design

Exp	Config	Purpose	Held fixed / disabled
Exp 1	`exp1_uncoordinated.yaml`	Uncoordinated baseline	No CTC, no MILP, no L-WSA
Exp 2	`exp2_opportunistic.yaml`	Natural whitespace only	No WiFi rescheduling
Exp 3	`exp3_ecc.yaml`	NAV-style ECC	Fixed ZigBee superframe
Exp 4	`exp4_cowic_full.yaml`	Full CoWiC-MAC	Soft MILP + L-WSA + sensing window
Exp 5	`exp5_rigid_vacate.yaml`	Hard WiFi vacate ablation	Hard MILP + fixed ZigBee superframe
Exp 6	`exp6_overhead_free.yaml`	Overhead-free variant	Soft MILP + L-WSA; no sensing window

Exp 6 also changes CTC from bidirectional to WiFi-to-ZigBee.

Parameter sweep dimensions:

Parameter	Values
D_{free}	31, 62, 93, 123 ms
T_{ZB}	1000, 2000, 5000 ms
Traffic	Traffic I and III by default; I–IV with `--traffic`
D_{sensing}	10, 20, 50, 100 ms
Seeds	5

Acceptance criterion: Exp 4 at $D_{\text{free}} = 31$ ms and $T_{ZB} = 1000$ ms must satisfy $G_z > 0.9$ and $PDR_w = 0$.

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9. Evaluation and Key Claims

Hypothesis mapping:

Hypothesis	Claim	Simulation support
H1	$G_z \geq 90\%$ under heavy load	Exp 4 with MILP + L-WSA
H2	$PDR_w = 0$	Deadline-aware MILP postponement
H3	Compare variant configurations	Exp 5 changes the scheduler and ZigBee superframe. Exp 6 removes sensing and uses one-way CTC.

Observed quick-sweep acceptance point (Exp 4, Traffic I, $D_{\text{free}} = 31$ ms, $T_{ZB} = 1000$ ms):

- $G_z = 0.929$
- $PDR_w = 0.0000$
- Result: PASS

This point meets the Issue #11 high-load acceptance rule. It does not by itself claim that every D_{free} value exceeds 0.9.

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10. Limitations and Future Work

Known limitations of this software-only study:

1. No PHY fading, capture, or SNR model
2. CTC is an error-rate message model, not waveform CTC
3. No multi-hop ZigBee routing
4. No IEEE 802.11be multi-link operation

Useful extensions:

1. Export traces into ns-3 for RF-level checks
2. Add multi-ZC demand aggregation
3. Study 802.11be multi-link coexistence with the same MILP / L-WSA split
4. Replace abstract PER with measured CTC traces

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11. References

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Base paper

Cheng, T., Li, S., and Pan, Y. "CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks." *Computer Networks*, 287, Art. 112479, 2026. DOI: 10.1016/j.comnet.2026.112479.